Dividing by a One-Digit Number

Partial quotients, long division, and estimating before you divide

Directions:

- Show the division, not just the answer — long division steps, or partial quotients written underneath each other and added at the end.
- Check every answer by multiplying back: (quotient) $\times$ (divisor) + (remainder) should rebuild the number you started with.
- Where a problem has no remainder, say so by writing the quotient alone.

1. Divide. $84 \div 4$

Answer: __________________

2. Divide. $96 \div 6$

Answer: __________________

3. Divide $258 \div 6$ using partial quotients. Start by taking out 40 groups of 6.

$6 \times 40 =$ _________ so $258-$ that $=$ _________

How many more groups of 6 are in what is left? _________

Add the partial quotients for the total.

Answer: __________________

4. Divide $475 \div 8$. There will be a remainder.

Quotient: ________ Remainder: ________

5. Estimate first, then divide $3{,}142 \div 6$.

(a) Round 3,142 to the nearest thousand. ________

(b) Use that to estimate the quotient. ________

(c) Divide exactly: quotient ________, remainder ________.

6. Divide. $2{,}568 \div 8$

Answer: ________________

7. Divide $618 \div 6$.

(a) What is the quotient? ________

(b) The quotient has a 0 in it. Explain why that zero has to be written.

8. Divide $5{,}207 \div 9$.

Quotient: _________ Remainder: _________

9. Estimate first, then divide $4{,}825 \div 5$.

(a) Round 4,825 to the nearest thousand. _________

(b) Use that to estimate the quotient. _________

(c) Now divide exactly. _________

10. Divide $4{,}872 \div 7$ using partial quotients. Start by taking out 600 groups of 7.

$7 \times 600 =$ _________ so $4{,}872-$ that $=$ _________

How many more groups of 7 are in what is left? _________

Add the partial quotients for the total.

Answer: __________________

11. A four-digit number is divided by a one-digit number. The number is 1,908 and the quotient is exactly 212, with no remainder. What is the divisor?

Answer: __________________

12. Divide 7,236 ÷ 4.

(a) What is the quotient? ________

(b) Explain how you could tell, before doing any dividing, that the quotient would have four digits.